IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

THEMBUMENZI
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Core Objective:
Predict first-stage recovery to estimate commercial launch margins.

Key Driver:
Payload mass inversely correlates with recovery success.

Model Result:
Classification algorithms achieved 83.3% test accuracy.

Introduction

The Commercial Dilemma

Cost Reduction: Reusable boosters lower launch costs dramatically.

Risk Analysis: Mission profiles dictate remaining fuel reserves.

Risk Analysis: Mission profiles dictate remaining fuel reserves.

$62M

Standard Falcon 9 Commercial Launch Cost

Section 1

Methodology



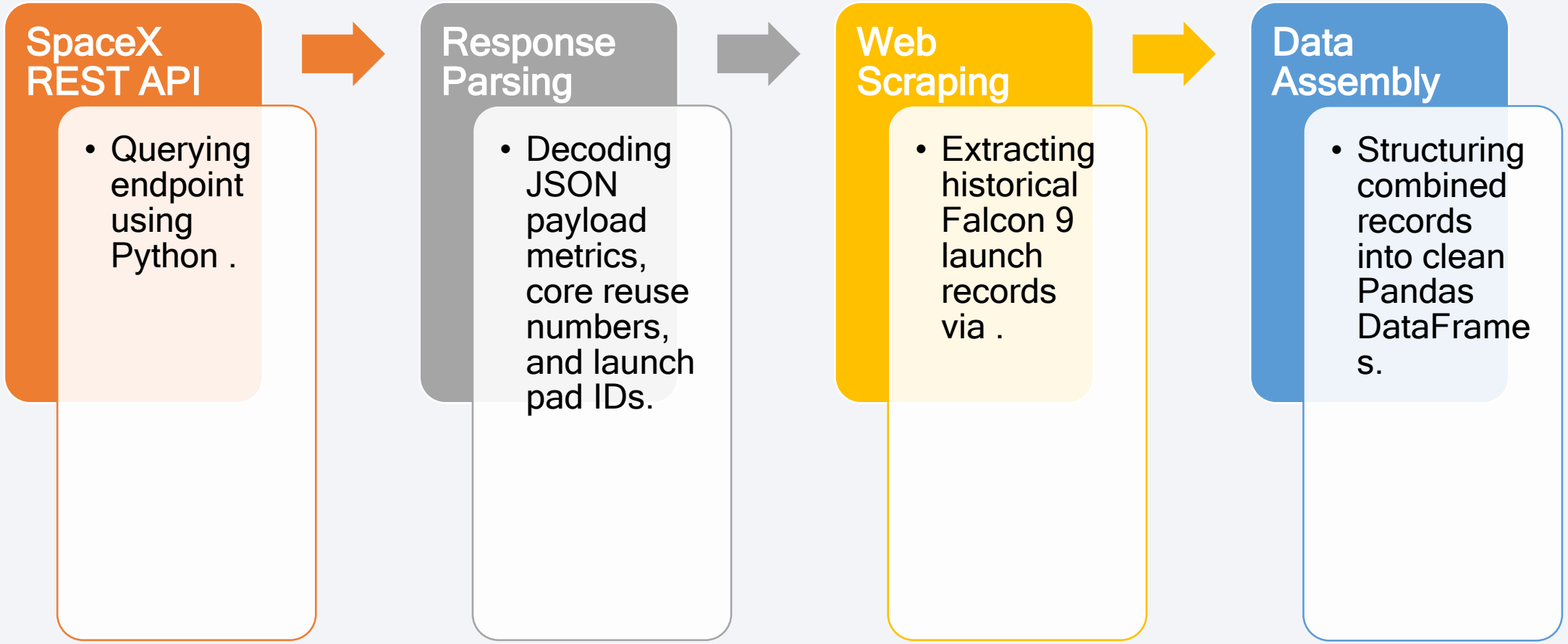
Methodology

Executive Summary

- Data collection methodology:
  - Gathered historical launch records via SpaceX REST API, web scraping, and SQL databases.
  - Perform data wrangling
  - Cleaned missing values, filtered records and engineered the target variable (class: binary landing success).
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Trained, optimized (GridSearchCV), and evaluated classification models (SVM, Decision Trees, KNN, Logistic Regression)

Data Collection:

Synthesizing structured REST API endpoints and unstructured web tables into a unified analytical dataset.

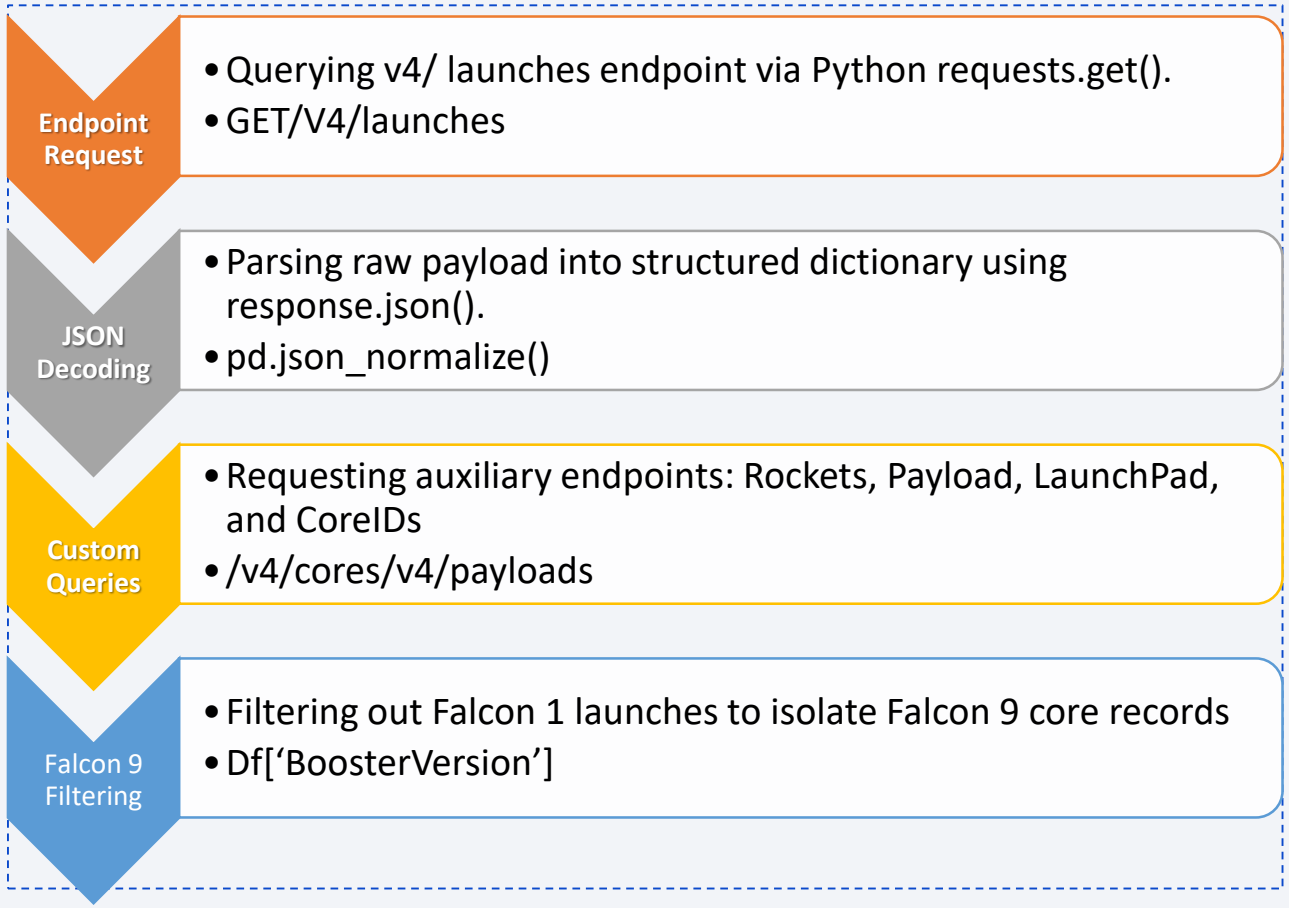


Data Collection - SpaceX API

Extracting launch metadata, core specifications, and target parameters via SpaceX API v4.

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

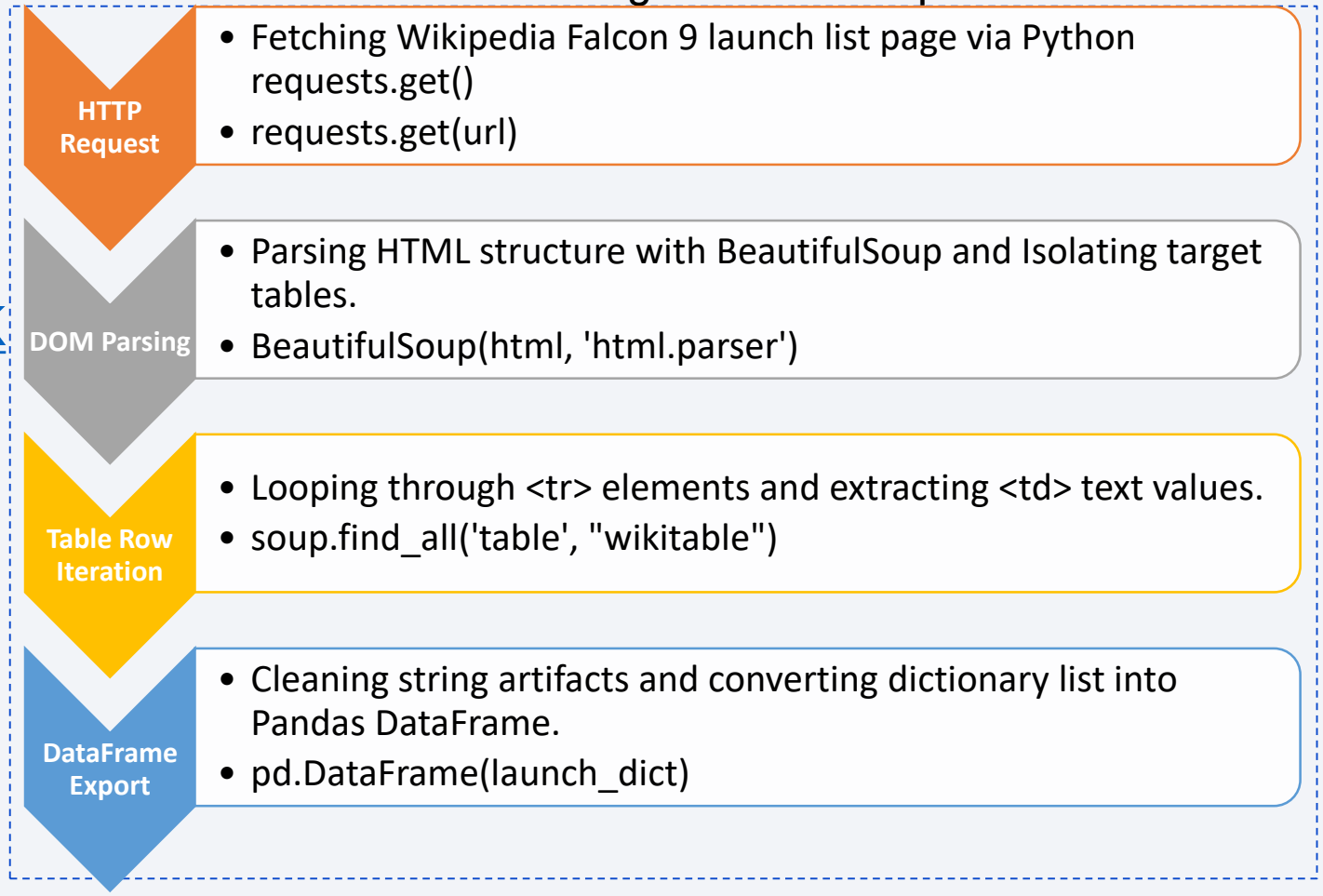
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/1jupyter-labs-spacex-data-collection-api_Pract1.ipynb



Data Collection - Scraping

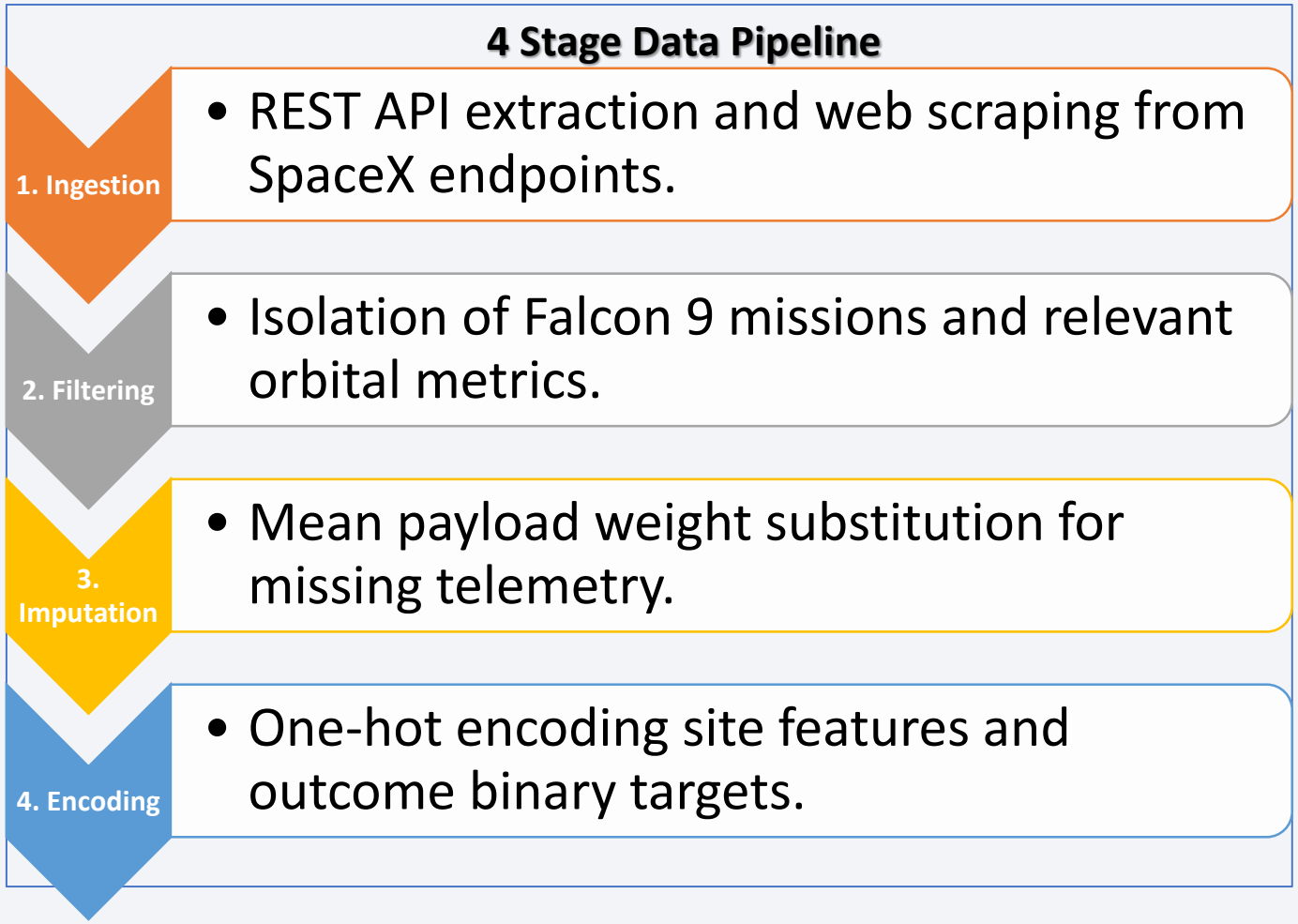
Extracting historical Falcon 9 launch records from HTML tables using BeautifulSoup4

- Present your web scraping process using key phrases and flowcharts
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/1jupyter-labs-webscraping_Pract2.ipynb



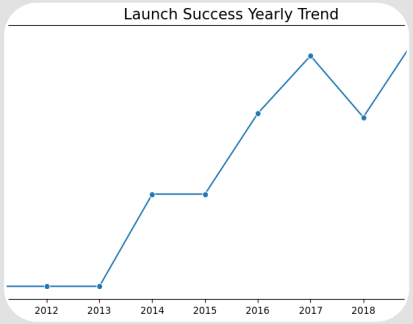
Data Wrangling

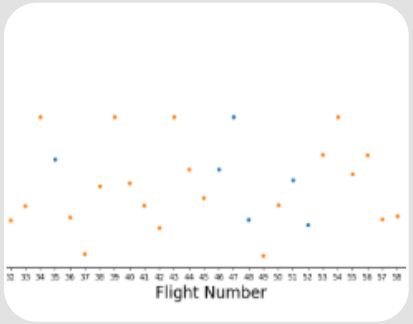
- Extracting, cleaning, and preparing launch telemetry and payload datasets for predictive modelling.
- Target Class Outcome Conversion: 67% Successful First-Stage Landing Rate in Wrangled Dataset
- Data Ready for Modelling: Cleaned dataset prepared for Logistic Regression, SVM, Decision Tree, and KNN classification.
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/2labs-jupyter-spacex-Data%20wrangling_Pract3.ipynb

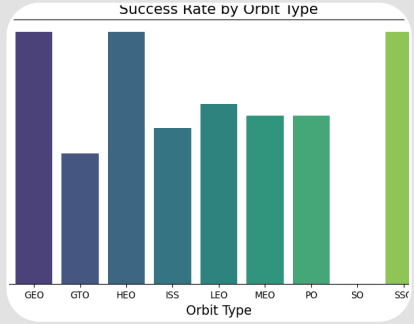


EDA with Data Visualization

- Key chart selections and analytical objectives for evaluating Falcon 9 landing success factors.
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/2edadataviz_PRACT5.ipynb







Line Chart: Tracks Yearly Trends. Plotted landing success rate over time (2010–2020) to visualize continuous improvement and learning curves across launch iterations.

Scatter Plots: **Feature Correlation & Clusters.** Analyzed Payload Mass vs. Flight Number and Launch Site vs. Flight Number to identify payload limits and site-specific outcome clusters.

Bar Chart: **Categorical Success Comparison.** Compared average landing success rates across target Orbit types (LEO, GTO, ISS, VLEO) to determine orbital insertion difficulty.

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/2 jupyter-labs-eda-sql-coursera_sqllite_PRACT4.ipynb

SQL Query Operations Executed

- **Unique Launch Sites:** Retrieved distinct names of launch locations using SELECT DISTINCT Launch_Site.
- **Pattern Matching (Cape Canaveral):** Filtered top records starting with 'CCA' using WHERE Launch_Site LIKE 'CCA%' LIMIT 5.
- **NASA CRS Total Payload:** Calculated total mass delivered to ISS using SUM(PAYLOAD_MASS__KG_) for customer NASA (CRS).
- **Booster Payload Averages:** Computed average payload mass specifically carried by version F9 v1.1 boosters.
- **First Ground Landing Milestone:** Found earliest date of successful ground pad landing using MIN(Date) function
- **Drone Ship Payload Range:** Listed boosters achieving drone ship success with payload mass between 4,000 kg and 6,000 kg.

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/3lab_jupyter_launch_site_location_PRACT6.ipynb

Custom Folium map visual elements used to evaluate launch site proximity, safety buffers, and logistical infrastructure.

Launch Site Markers & Clusters

- **Purpose: Site Isolation & Success Rates**
- Added color-coded MarkerCluster objects (green for success, red for failure) to quickly visualize historical launch outcomes geographically across pads

Circle Overlays

- **Purpose: Proximity & Buffer Zones**
- Drawn Circle overlays around launch pads to assess proximity to coastlines, residential cities, and high-density population areas for hazard management.

Distance Lines & PolyLines

- **Purpose: Infrastructure Calculation**
- Rendered PolyLine paths connected to coordinate markers (using Haversine distance) to quantify site distances to railways, highways, and coastlines

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/4spacex-dash-app.py

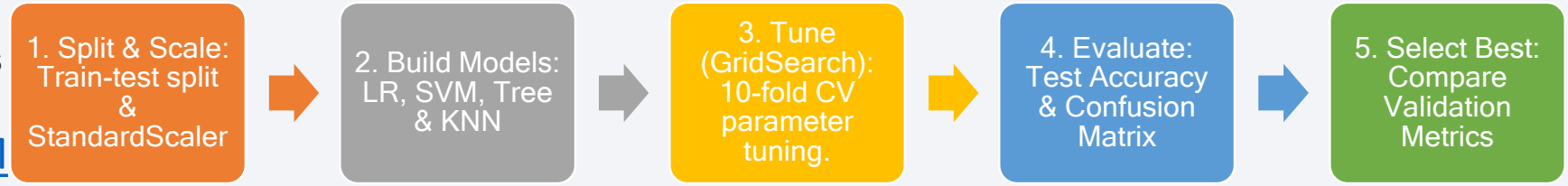
Built a dynamic web application enabling interactive site filtering and payload analysis for launch outcome evaluation.

Launch Site Dropdown	Payload Range Slide	Success Pie Chart	Payload Scatter Plot
• **Interaction: Site Selection** • Allows toggling between 'All Sites' and specific launch complexes (e.g., KSC LC-39A, CCAFS SLC-40) to dynamically trigger callback updates across charts.	• **Interaction: Mass Range Filtering** • Enables users to filter mission data across payload ranges (0-10,000+ kg) to evaluate landing reliability under varied weight constraints.	• **Plot: Total / Site Success Ratio** • Renders overall landing contribution when 'All Sites' is selected, or site-specific success vs. failure proportions when a single site is chosen.	• **Plot: Payload vs. Outcome** • Displays correlation between Payload Mass (kg) and Launch Outcome (0/1), color-coded by Booster Version category.

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- https://github.com/MenziNKhoza/Data-Science-Capstone-Project/blob/main/4notebooks_Final%20Project%20Submission_SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Built and tuned binary classifiers using StandardScaler feature preprocessing, GridSearchCV hyperparameter optimization, and 10-fold cross-validation



- **Model Construction:** Trained Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbours on normalized feature matrices.
- **Evaluation Strategy:** Assessed holdout test accuracy and generated confusion matrices to measure false positive/negative rates across models.
- **Optimization & Best Model:** All models achieved 83.3% test accuracy; Decision Tree / Logistic Regression selected as top performers due to simplicity and generalization.

Results

Summary of core insights derived from EDA and final predictive performance across all classification models.

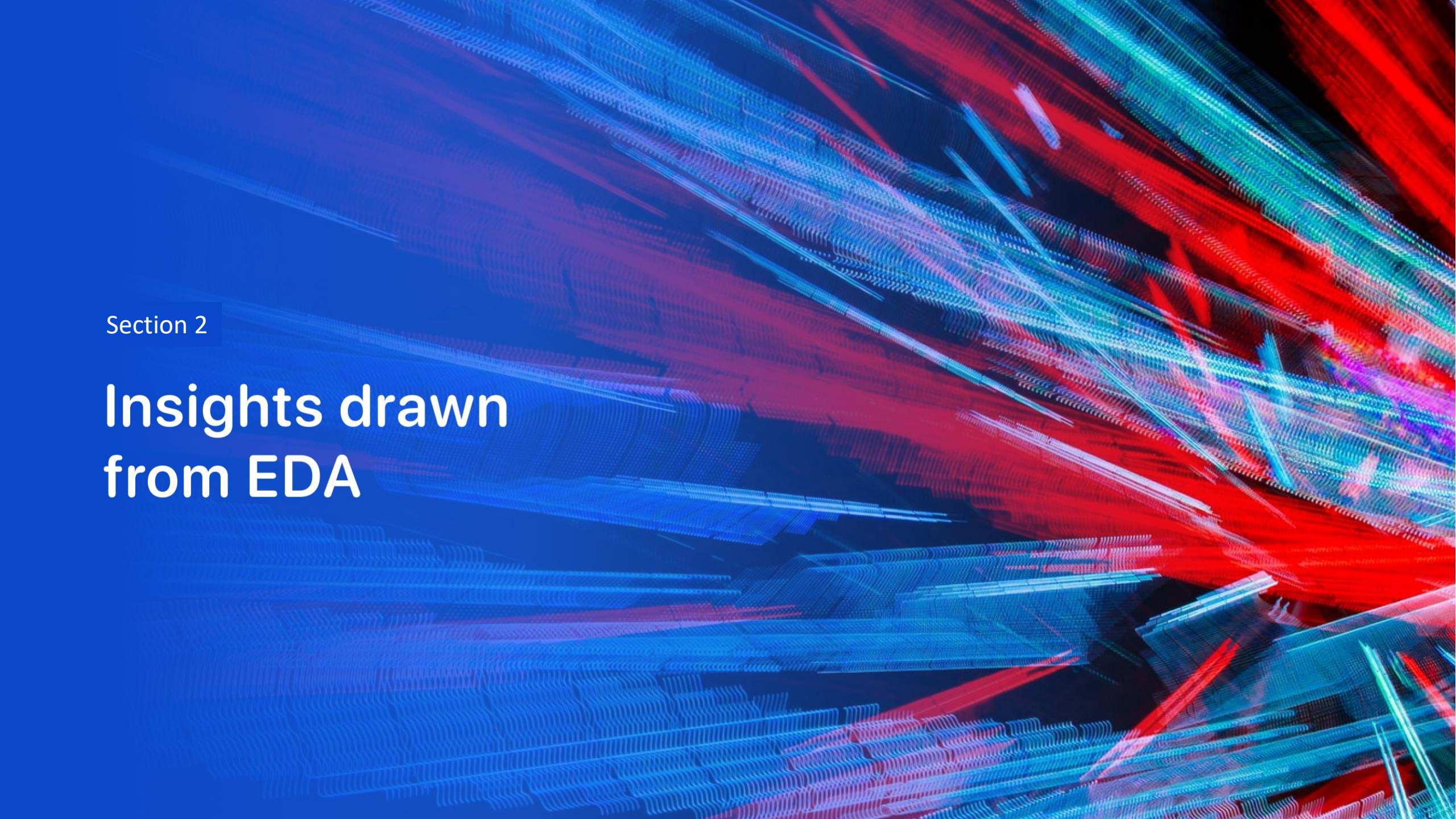
- Exploratory data analysis results
- Predictive analysis results
- Interactive analytics demo in screenshots

EDA Insights

- **Site Success**: KSC LC-39A demonstrated the highest overall launch success rate (~76%).
- **Payload Mass Effect:** Payloads between 2,000 kg and 5,000 kg yielded higher first-stage recovery rates.
- **Orbit Performance:** Orbits like HEO, LEO, and GEO achieved 100% success rates, while GTO averaged ~50%.
- **Trajectory over Time**: Landing success grew consistently post-2013 due to Block 5 booster refinements.

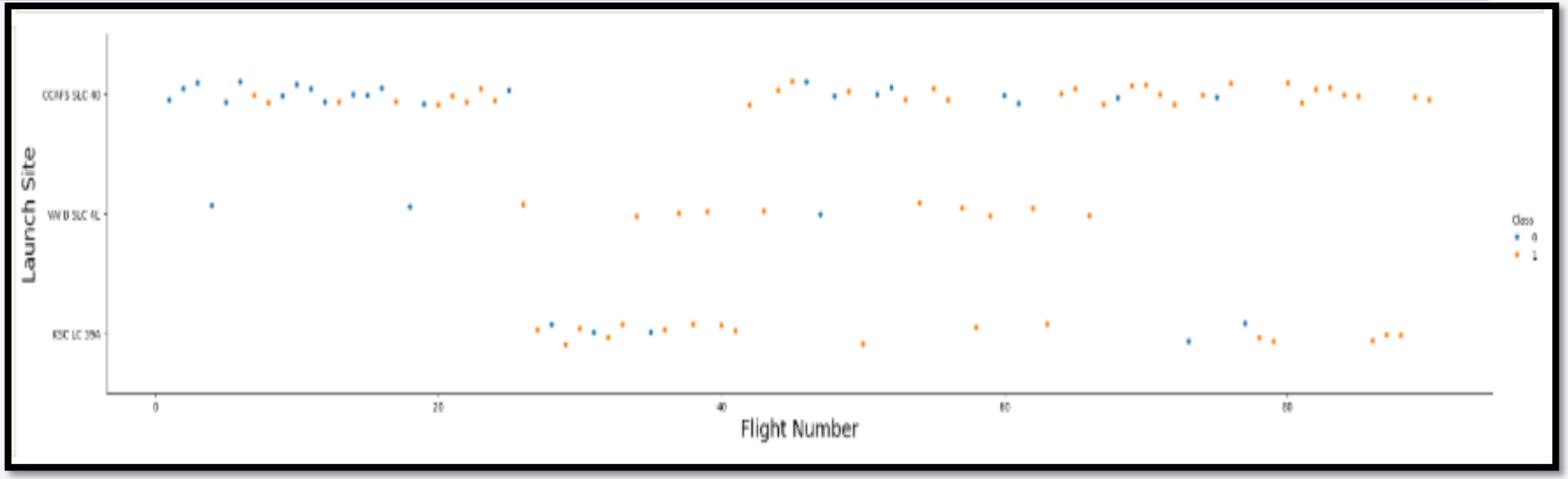
Predictive Insights

- **Test Accuracy**: All models (LR, SVM, Decision Tree, KNN) achieved a benchmark 83.33% test accuracy.
- **Best Hyperparameters**: Decision Tree and Logistic Regression performed best post-GridSearchCV tuning.
- **Confusion Matrix**: Models accurately predicted landing outcomes with low false-positive rates on unseen data.
- **Key Predictors**: Payload mass, orbit type, and launch site served as primary predictive drivers.


Section 2
Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site
- Show the screenshot of the scatter plot with explanations

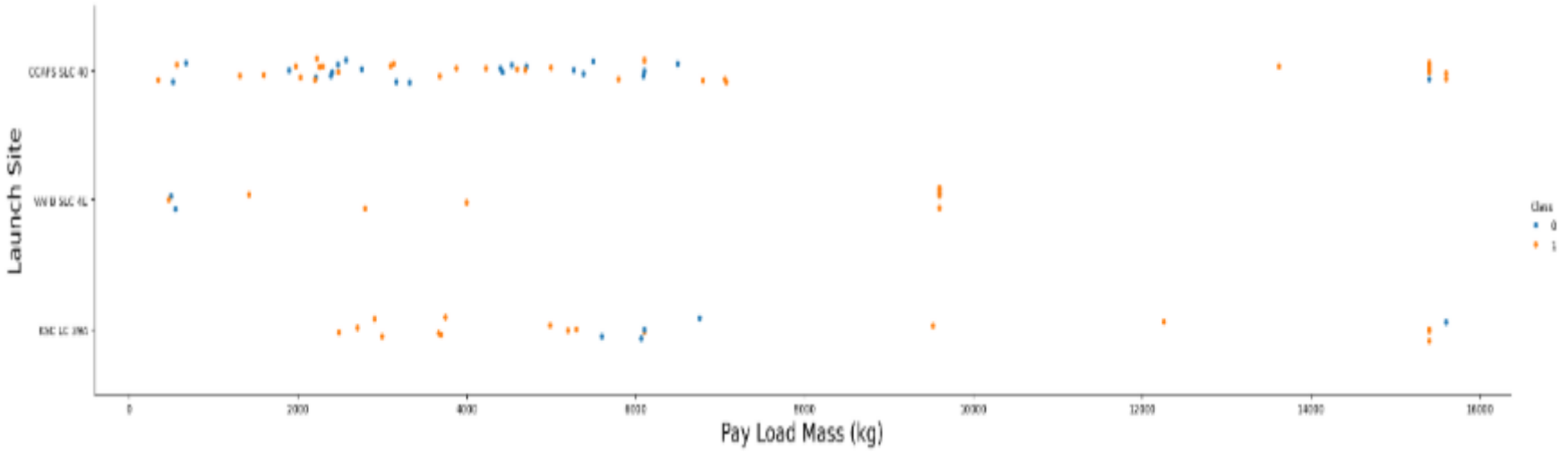


Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots.

The relationship between FlightNumber and Launch is NOT LINEAR. It is scattered with some outliers/noise within the LaunchSite: VAFB SLC 4E. As the FlightNumber increases (tightly), so does the Launchsite increase, resting at LaunchSite: CCAFS SLC 40 but then dips between FlightNumber 20-40 (by LaunchSite: KSC LC 39A) then spikes up again to LaunchSite: CCAFS SLC 40 (more spaced-out) with more outliers /noise within the LaunchSite: VAFB SLC 4E and LaunchSite: KSC LC 39A.

Payload vs. Launch Site

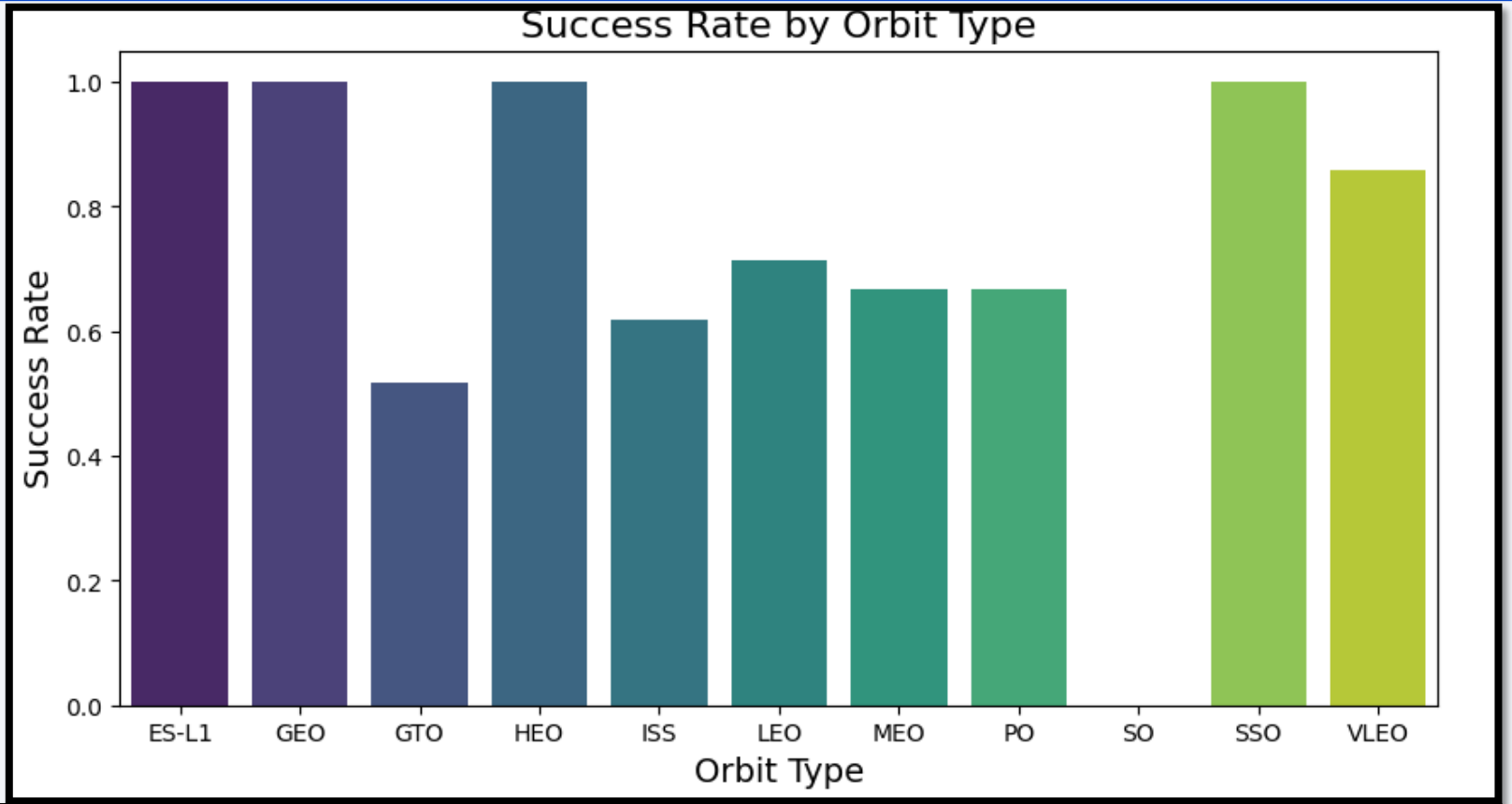
- Show a scatter plot of Payload vs. Launch Site
- Show the screenshot of the scatter plot with explanations



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type

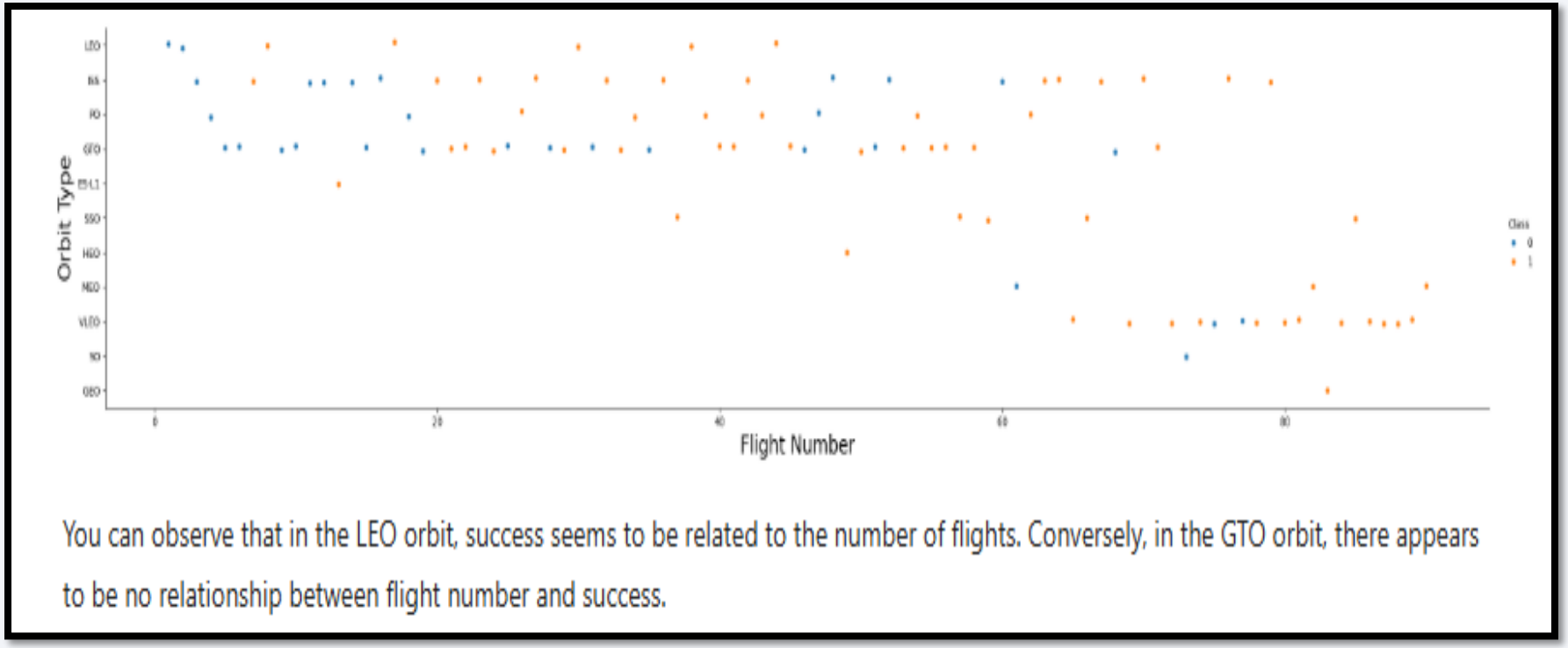
- Show a bar chart for the success rate of each orbit type
- Show the screenshot of the scatter plot with explanations



Orbit HEO: Appears to stand higher than ES-L1, GEO and SSO by a point ~1.1/1.2. Orbit Type: ES-L1, GEO and SSO have the same success rate of 1.0 Orbit VLEO: has a success rate of +/-0.9

Flight Number vs. Orbit Type

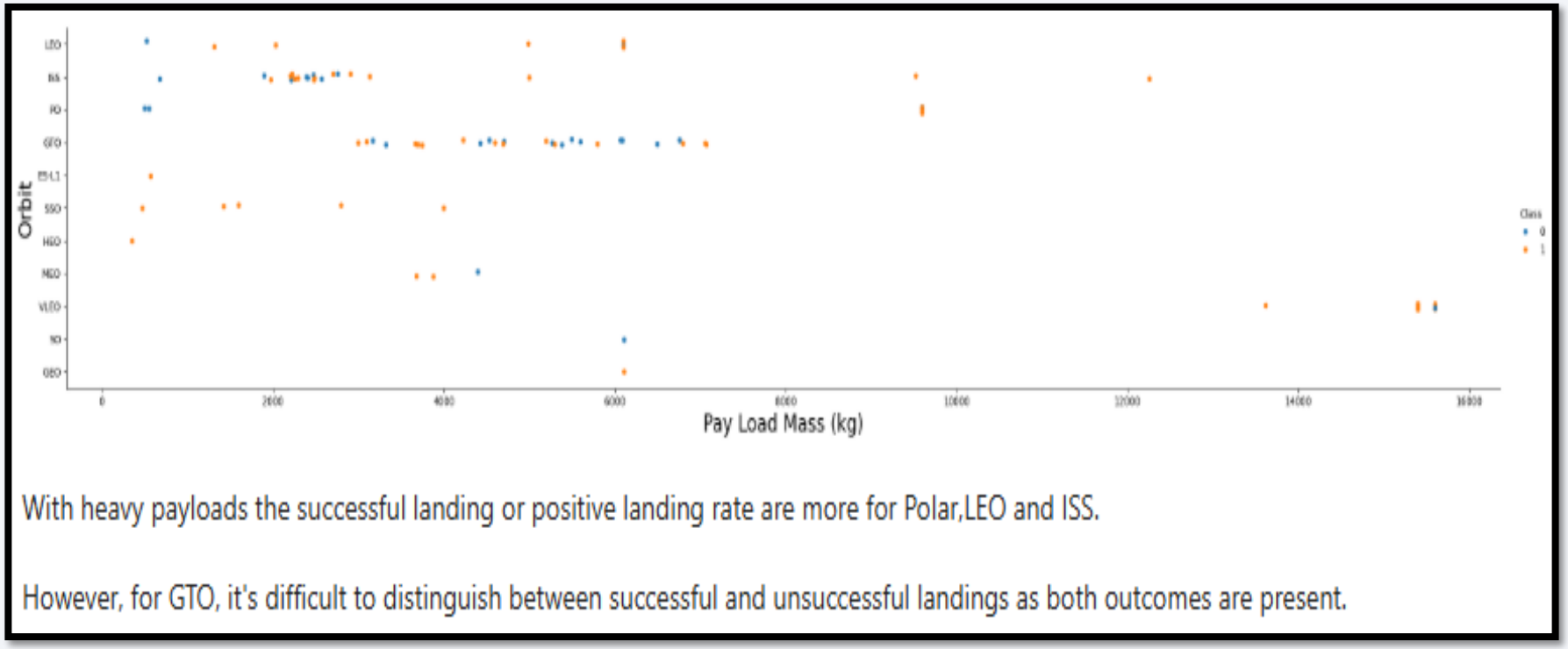
- Show a scatter point of Flight number vs. Orbit type
- Show the screenshot of the scatter plot with explanations



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

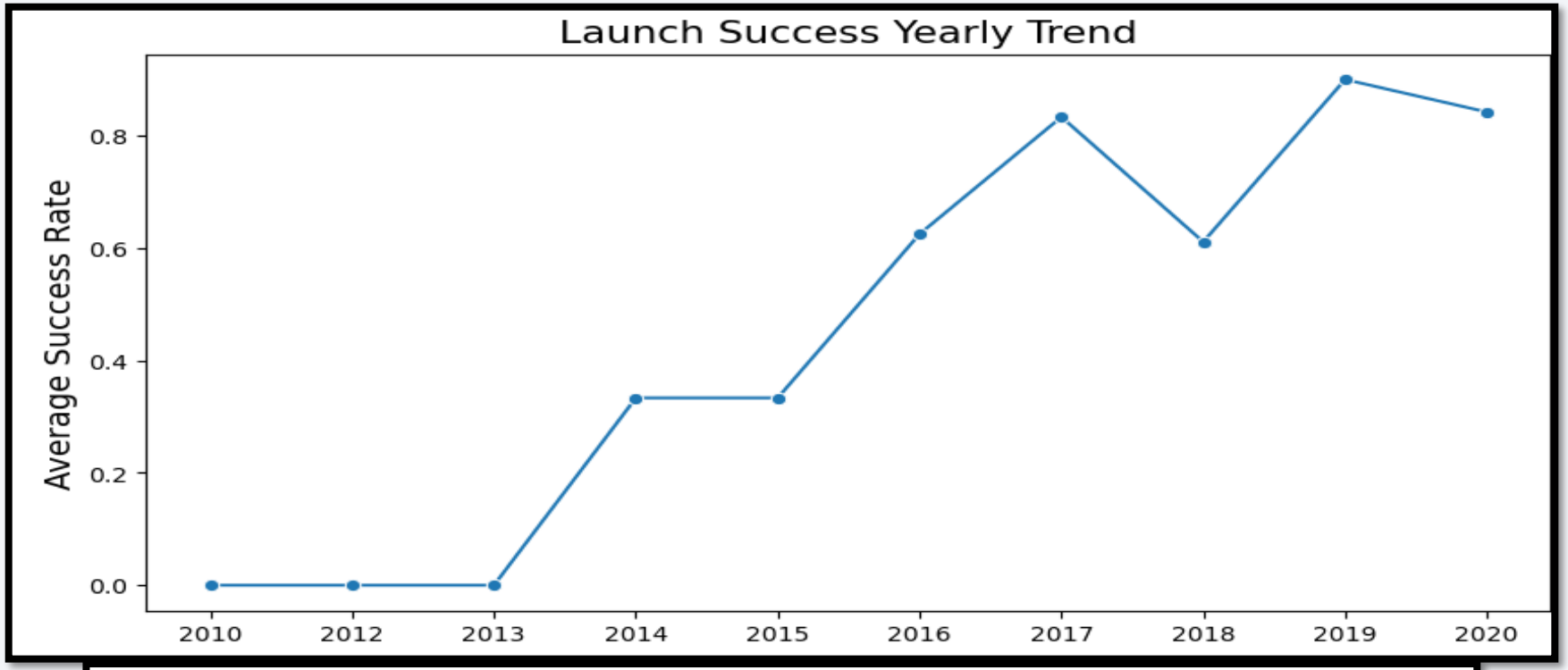
Payload vs. Orbit Type

- Show a scatter point of Payload vs. Orbit Type
- Show the screenshot of the scatter plot with explanations



Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Show the screenshot of the scatter plot with explanations



you can observe that the sucess rate since 2013 kept increasing till 2020

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

```
%%sql
SELECT DISTINCT “Launch_Site” FROM SPACEXTBL;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

This query retrieves a list of all **unique (distinct) launch sites** from the SPACEXTBL table, ensuring that each launch facility name appears only once

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Present your query result with a short explanation here

```
%%sql
            SELECT*
FROM SPACEXTBL
WHERE “Launch_Site” LIKE ‘CCA%’
LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

This query retrieves the first 5 records from the dataset for any launch sites starting with the prefix 'CCA' (such as Cape Canaveral Air Force Station facilities like CCAFS SLC-40), displaying all columns for those specific missions

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here

```
%%sql
SELECT SUM("PAYLOAD_MASS__KG_")
FROM SPACEXTBL
WHERE "CUSTOMER" = 'NASA (CRS)';
```

SUM("PAYLOAD_MASS__KG_")
45596

This query calculates the total cumulative payload mass (in kilograms) sent to space specifically for missions contracted by the customer **NASA (CRS)**, giving the total cargo weight delivered by SpaceX for those commercial resupply flights.

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

```
%%sql
SELECT AVG("PAYLOAD_MASS__KG_")
FROM SPACEXTBL
WHERE "Booster_Version" = 'F9 v1.1' ;
```

AVG("PAYLOAD_MASS__KG_")
2928.4

This query calculates the average payload mass (in kilograms) across all SpaceX missions that utilized the **F9 v1.1** booster version, providing insight into the typical cargo weight capacity for that specific rocket configuration.

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

```
%%sql
SELECT MIN(Date)
FROM SPACEXTBL
WHERE “Landing_Outcome” LIKE ‘%Success (ground pad)%’ ;
```

MIN(Date)
2015-12-22

This query identifies the earliest date in the dataset where a booster achieved a successful landing on a ground pad, marking the historic date of SpaceX's very first successful ground landing

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Present your query result with a short explanation here

```
%%sql
SELECT DISTINCT “Booster_Version”
FROM SPACEXTBL
WHERE “Landing_Outcome” LIKE ‘%Success (drone ship)%’
            AND “PAYLOAD_MASS__KG_” > 4000
            AND “PAYLOAD_MASS__KG_”  <6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

This query retrieves the unique booster versions that successfully landed on a drone ship while carrying a payload mass strictly between 4,000 kg and 6,000 kg, helping identify which rocket configurations successfully handled mid-range cargo weights.

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

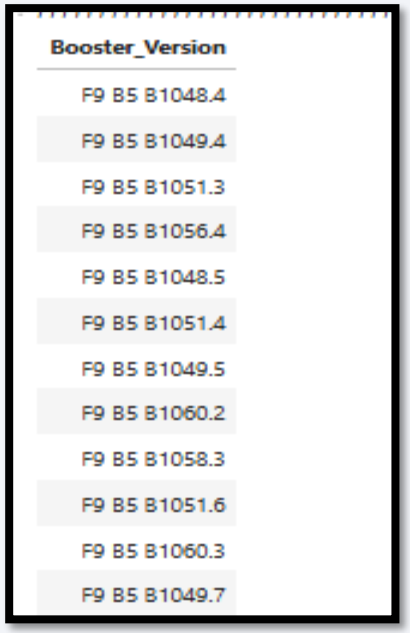
```
%%sql
SELECT “Mission_Outcome”, COUNT(*) AS “Total_Count”
FROM SPACEXTBL
GROUP BY “Mission_Outcome”;
```

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

This query groups all missions by their outcome status (such as successful or failed flights) and counts the total number of launches in each category, providing a breakdown of overall mission success rates across the dataset.

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here



Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

```
%%sql
SELECT DISTINCT “Booster_Version”
FROM SPACEXTBL
WHERE “PAYLOAD_MASS__KG_” = (
            SELECT MAX(“PAYLOAD_MASS__KG_”)
            FROM SPACEXTBL
);
```

This query uses a subquery to find the maximum payload mass recorded across the entire dataset and then identifies the specific booster version(s) that were used to carry that heaviest cargo load into space.

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Present your query result with a short explanation here

```
%%sql
SELECT substr(Date, 6 , 2) AS “Month”,
            “Landing_Outcome”,
            “Booster_Version”,
FROM SPACEXTBL
WHERE substr(Date, 1, 4) = ‘2015’
AND “Landing_Outcome” LIKE ‘%Failure (drone ship)%’;
```

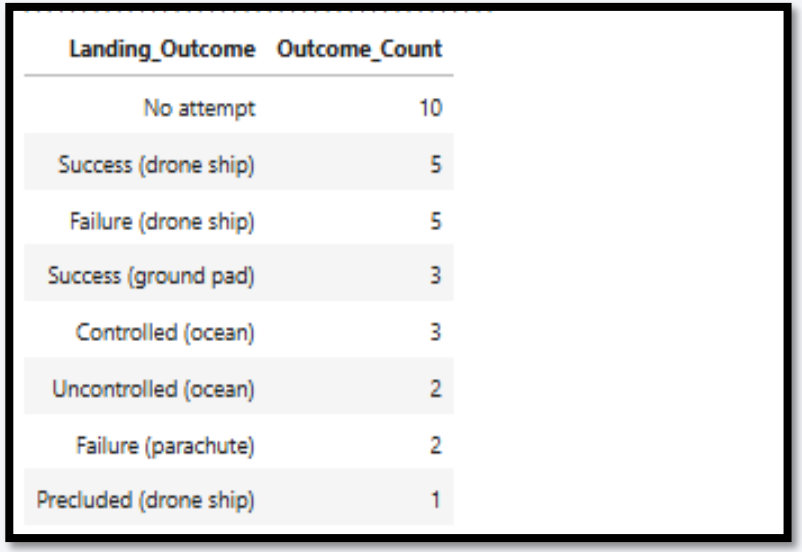
Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

This query extracts the month, landing outcome, and booster version for missions that took place in the year 2015 and specifically resulted in a drone ship landing failure, helping track and analyze unsuccessful ocean landing attempts during that year.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here

```
%%sql
SELECT “Landing_Outcome”, COUNT(*) AS “Outcome_Count”
FROM SPACEXTBL
WHERE “Date” BETWEEN ‘2010-06-04’ AND ‘2017-03-20’
GROUP BY “Landing_Outcome”
ORDER BY “Outcome_Count” DESC;
```



Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

This query counts the total occurrences of each unique landing outcome between June 4, 2010, and March 20, 2017, and sorts them in descending order (highest frequency first), providing a ranked breakdown of booster landing success and failure types during that specific historical date range.


Section 3
Launch Sites
Proximities Analysis

Folium Map: Main SpaceX Launch Geo Coordinates

- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map
- Explain the important elements and findings on the screenshot

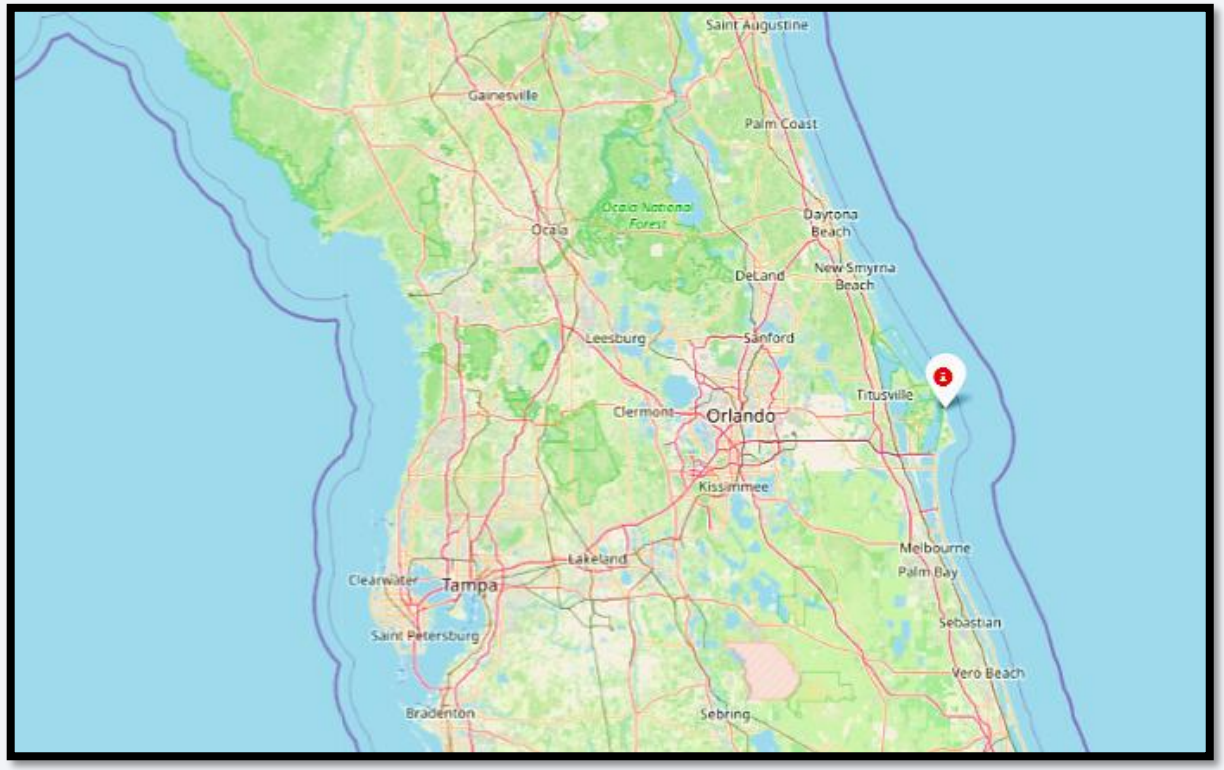


LAUNCH SITES	LAT	LONG
CCAFS LC-40	28,562302	-80,577356
CCAFS SLC-40	28,563197	-80,576820
KSC LC-39A	28,573255	-80,646895
VAFB SLC-4E	34,632834	-120,610745

This screenshot represents is the outcome of the main dataframe to retain only the necessary columns (Launch Site, Lat, Long, and class), and then groups the data by each launch site to extract its unique geographic coordinates. This creates a clean summary dataframe (launch_sites_df) used to plot the core launch site markers on the Folium interactive map. **Launch sites: indicated on USA map with orange circles.**

Geographic Mapping & Outcome Clustering in Folium

- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Explain the important elements and findings on the screenshot

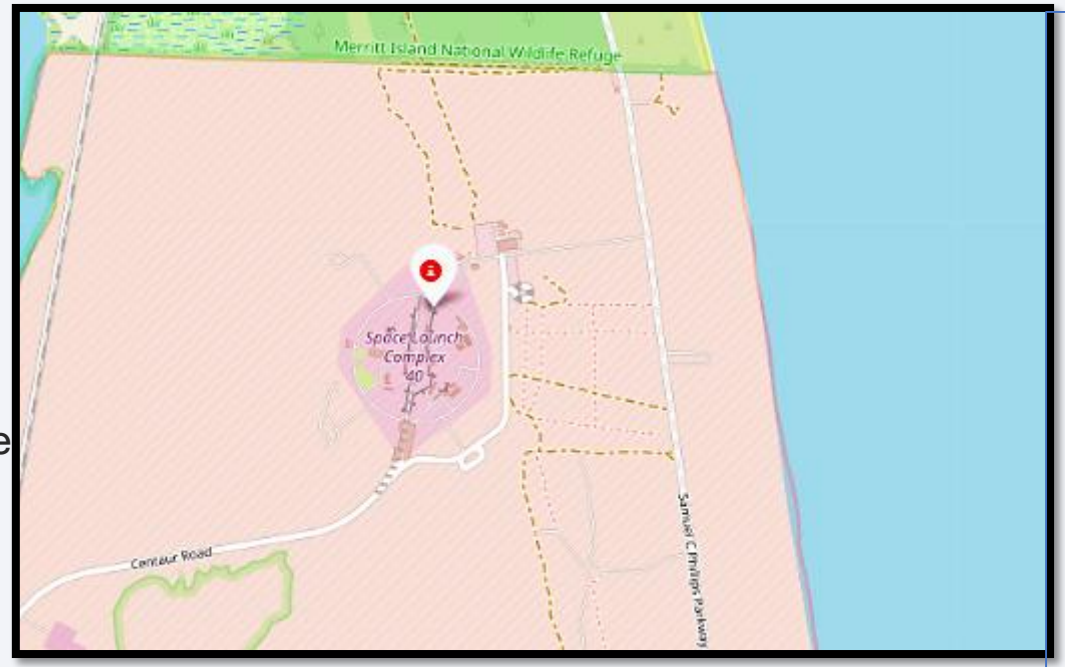


1. **Centralized Geographic Mapping:** The map initializes with a regional view (zoom_start=5) anchored near NASA Kennedy Space Center, providing a clear overview of all SpaceX launch locations across different regions.
2. **Visual Outcome Color-Coding:** A conditional list comprehension (green for success 1, red for failure 0) automatically tags every mission. On the screenshot, this instantly communicates the **success versus failure ratio** for each launch site at a glance.

3. **Marker Clustering for High-Density Data:** The MarkerCluster() tool automatically groups overlapping or nearby launch markers into interactive count bubbles when zoomed out. This keeps the interface clean and prevents visual clutter while allowing viewers to drill down into specific launch locations as they zoom in.

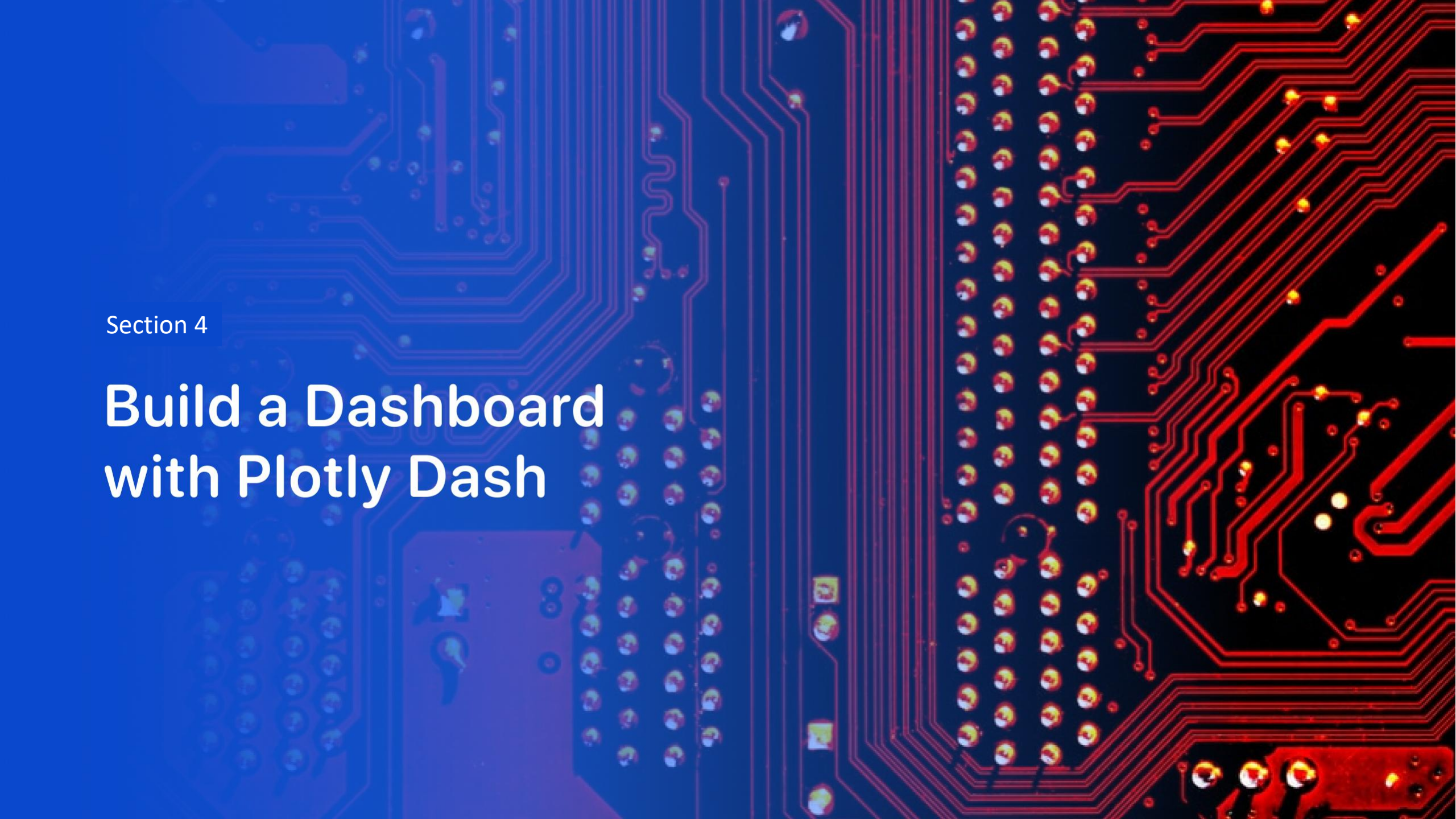
Proximity Analysis: Launch Site Infrastructure & Geographic Markers

- Replace <Folium map screenshot 3> title with an appropriate title
- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- Explain the important elements and findings on the screenshot



- **Core Insight:** Demonstrates how SpaceX balances public safety (coastal positioning) with logistical accessibility (direct rail/road access).

- **Distance Measurements:** Polylines and custom markers dynamically display calculated distances (in km) between the launch pad and key landmarks.
- **Coastline Proximity:** Visualizes launch trajectories over water, highlighting safety protocols that minimize risk over populated areas.
- **Highways & Railways:** Identifies critical transport corridors required for heavy booster logistics, propellants, and equipment.


Section 4
Build a Dashboard with Plotly Dash

Launch Success Distribution Across All Launch Sites

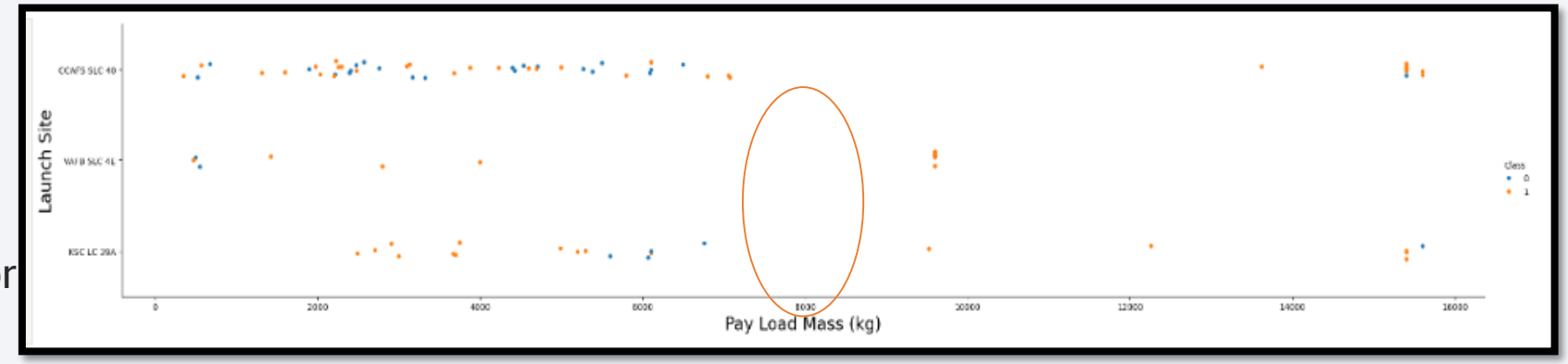
- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

Launch Success vs. Failure Ratio for Top-Performing Site

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot

Correlation Between Payload Mass and Launch Success Across All Sites

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.



Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000)- highlighted with a orange oval.

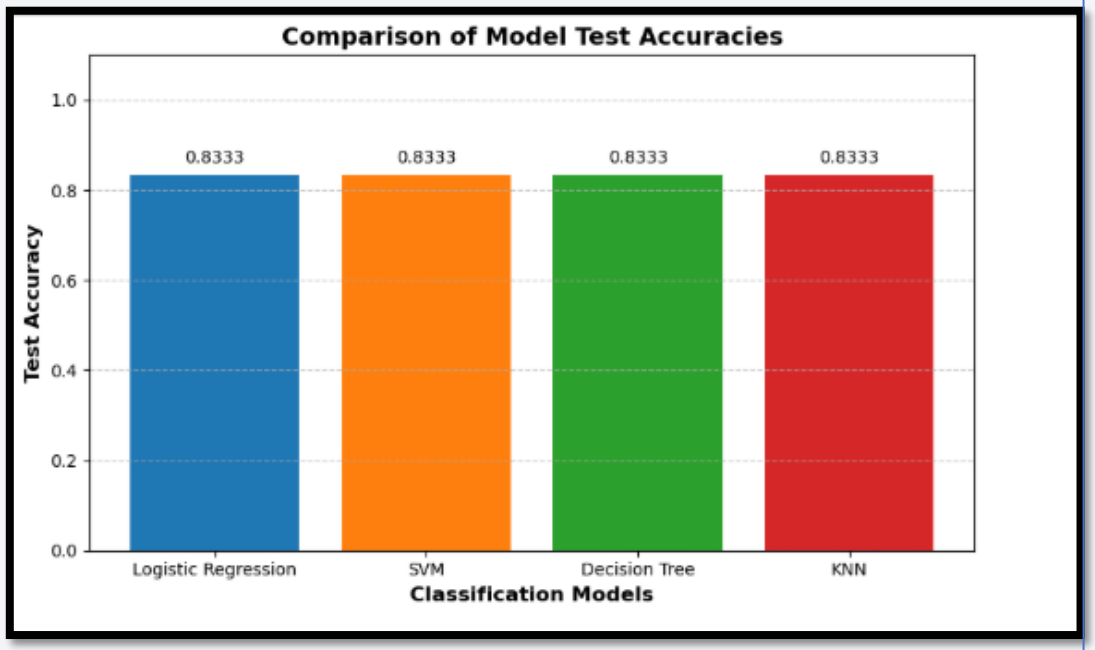
Section 5

Predictive Analysis (Classification)



Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart
- Find which model has the highest classification accuracy

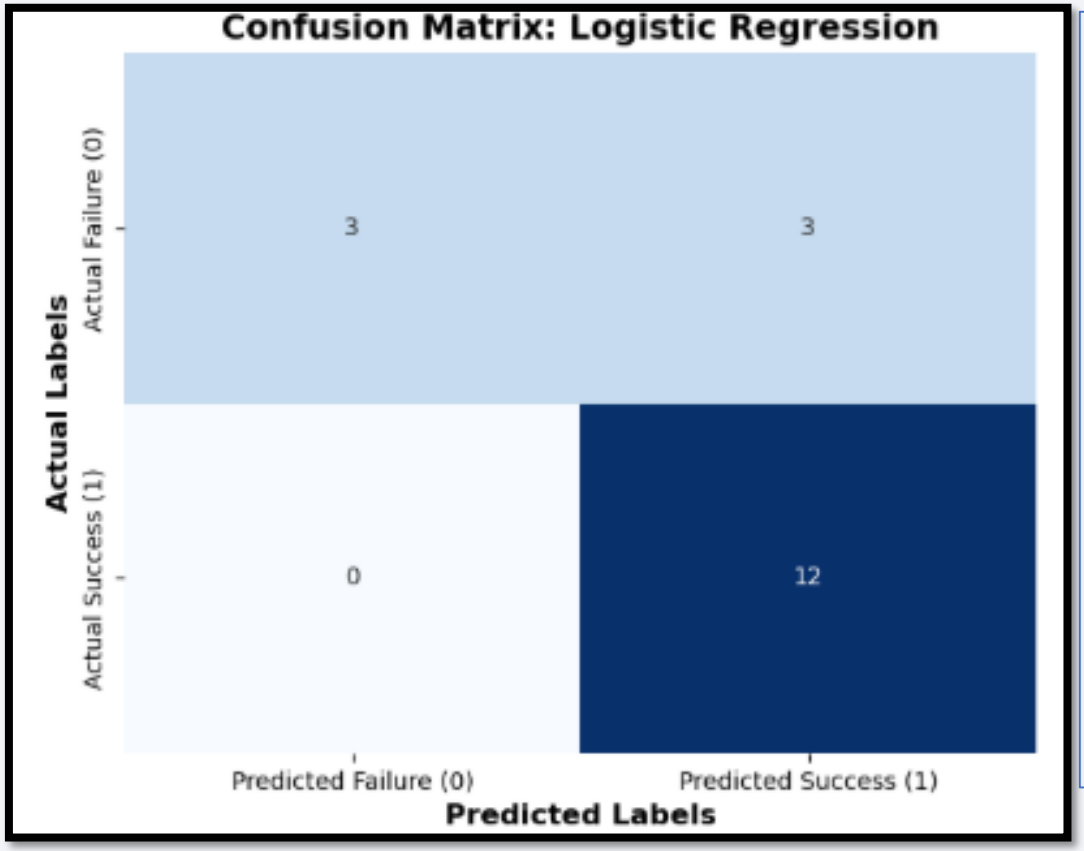


Highest Classification Accuracy: **83.33%**
All four classification algorithms achieved identical test set accuracy rates of **83.33%** after hyperparameter tuning.

Evaluation Takeaway
Decision Tree and Logistic Regression are favoured for deployment due to lower computational overhead and direct feature interpretability

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation



100% Sensitivity (Recall)
The model accurately identified all 12 successful landings without producing any false negative errors.

False Positive Tendency
The 3 misclassified test samples were failed landings predicted as successful, showing a slight optimistic bias toward booster recovery.

Conclusions

1. **Data analysis & SQL Insights**

Exploratory analysis identified success rates increasing significantly over time, with heavy payload capacity boosters achieving the highest operational consistency.

2. **Geographic & Safety Strategy**

Folium map visualizations confirmed that launch pads prioritize coastal positioning to ensure trajectory safety over water while retaining nearby rail/highway logistics access.

3. **Predictive Machine Learning**

Classification algorithms achieved **83.33% accuracy** in forecasting landing outcomes, with perfect detection performance for successful booster recoveries.

4. **Business Impact & Savings**

Predictive landing analytics allow commercial launch operators to estimate reusability risks accurately and optimize bid pricing against competing launch providers.

Appendix A: **Data Pipeline & Feature Engineering**

Wrangling & Binary Class Encoding

Target Extraction: Filtered LandingOutcome into binary Class column (1 = Successful drone ship/land touch down, 0 = Uncontrolled ocean crash/failure).

Missing Value Imputation: Replaced missing PayloadMass values with mean average (2,928 kg).

Dataset Split: Stratified 80/20 Train-Test split via train_test_split(X, Y, test_size=0.2, random_state=2).

One-Hot Encoding & Scaling

One-Hot Encoding: Converted categorical variables (Orbit, LaunchSite, LandingPad, Serial) using pd.get_dummies(), yielding 83 total features.

Feature Normalization: Applied StandardScaler().fit_transform(X) to standardize numerical features prior to model fitting.

Representative Database & Extraction

```
SELECT Launch_Site, COUNT(*) AS Total_Launches, SUM(CASE WHEN Landing_Outcome LIKE '%Success%' THEN 1 ELSE 0 END) AS Successes,
 ROUND(AVG(Payload_Mass_kg_), 2) AS Avg_Payload_kg
 FROM SPACEXTBL
 WHERE Payload_Mass_kg_ BETWEEN 2000 AND 6000
 GROUP BY Launch_Site ORDER BY Total_Launches DESC;
```

Appendix B: GridSearchCV Hyperparameters & Stack

Tuned Hyperparameters (GridSearchCV, cv =10)

Model	Parameter Grid Tested	Best Config
LogReg	C: np.logspace(-3,3,7), penalty: ['l2']	C = 0.01
SVM	C: np.logspace(-3,3,5), kernel: ['linear','rbf','sigmoid']	C = 1.0, sigmoid
Tree	criterion: ['gini','entropy'], max_depth: [2,4,6,8,10]	gini, depth = 4
KNN	n_neighbors: [1..10], algorithm: ['auto','ball_tree']	n_neighbors = 10

Technical Stack

Python 3.x (Pandas, NumPy, Scikit-learn)
Database: SQLite / IBM DB2 SQL
Dashboarding: Plotly Dash
Geospatial: Folium (MarkerCluster)

Data Artefacts

- Raw REST API JSON Response
- Scraping Wiki Landing Records
- Feature Matrix (90 rows × 83 cols)
- Folium Interactive HTML Map

Thank you!

